Rajesh Pandit

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Education

Bachelor of Technology (B.Tech) – Computer Science & Engineering
Sri Venkateswara College of Engineering, Chittoor
Expected Graduation: July 2028 CGPA: 7.74
Relevant Coursework: Data Structures, Operating Systems, DBMS, OOP

Projects

NexusChat – Distributed Real-Time Chat System Live App — Project Link — Demo
Node.js • Socket.io • MongoDB • System Design • RabbitMQ • Redis • NGINX

- **Distributed Real-Time Architecture:** Built a horizontally scalable chat backend using **Node.js + Socket.IO across 5 instances**, validated with **k6 at 5,000 concurrent WebSocket VUs** and **104K+ completed sessions**.
- **High-Throughput Event Pipeline:** Designed **Redis Pub/Sub** across 4 channels for cross-server message, typing, read, and presence events; load tests processed **2.97M+ WebSocket messages at 14.5K+ msgs/sec** with **˜14ms p95 send latency**.
- **Reliability & Load Validation:** Stress-tested real-time delivery to **2,701 concurrent users**, achieving **100% message delivery and 0 errors** in production-equivalent conditions.
- **Auth Pipeline:** Implemented JWT + email verification and bcrypt hashing, load-tested via **k6 at 200 concurrent VUs** — sustaining **5,700+ HTTP requests** at **98% login success** and **3.1ms median response time**.
- **Production Reliability:** Engineered SIGTERM graceful shutdown — broadcasting offline presence, flushing Redis mappings, and updating MongoDB `lastSeen` — ensuring **zero ghost-online users** across restarts.
- **Async Email Processing:** Moved email sending to **RabbitMQ** with retry backoff, DLQ, and Redis idempotency, reducing register latency by **74%** (**921 ms → 240 ms**).

LibraTech – Advanced Library Management System Project Link — Demo
Node.js, Express.js, MongoDB, JWT, HTML, CSS, JavaScript

- Architected and developed a full-stack library management system handling **users, books, issue/return workflows, fines, and analytics**.
- Designed secure **RESTful APIs** with **JWT-based authentication** and **role-based access control** (Admin / Student).
- Implemented **normalized MongoDB schemas** using Mongoose to ensure data integrity and efficient querying.
- Built **responsive dashboards** with real-time statistics for issued books and overdue returns.
- Automated **fine calculation logic** and optimized CRUD operations to improve system reliability.

Technical Skills

Languages: C++, JavaScript, Python
Backend: Node.js, Express.js
Databases: MongoDB, Redis, PostgreSQL
Infrastructure: Docker, Nginx, Linux
Messaging & Real-Time: RabbitMQ, Socket.IO, Redis Pub/Sub
Tools: Git, GitHub
Fundamentals: DSA, OOP, Operating Systems

Achievements

- Secured **1st Place** in a **Debugging Competition** at iTechnofest 2K25, held at SVCET, competing against **M.Tech, MCA, B.tech, and BCA students from multiple colleges**. Link
- Solved **600+ problems on LeetCode** across Trees, Graphs, DP, and Hashing — **Contest Rating: 1804**.